2025

MATHEMATICS

(Major–I)

Paper : MAT4104C

(Abstract Algebra)

Full Marks : 60

Time : 2½ hours

The figures in the margin indicate full marks for the questions.

1. Answer the following questions : $1 \times 7 = 7$

   (a) Is the set S of positive irrational numbers together with '1' under multiplication a group? Justify.

   (b) The set $\{5, 15, 25, 35\}$ is a group under multiplication modulo 40. Find the inverse of 25.

   (c) What is the order of the group $U(2025)$ under multiplication modulo 2025?

   (d) What is the order of the permutation $(124)(235)$ in S_5?

Contd.

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(e) Define external direct product of groups.

(f) Give an example of non-commutative group without unity.

(g) Is $2\mathbb{Z} \cup 3\mathbb{Z}$ a subring of $\mathbb{Z}$? Justify.

2. Answer the following questions : $2 \times 4 = 8$

(a) Find the inverse of the element

$$\begin{bmatrix} 2 & 6 \\ 3 & 5 \end{bmatrix} \text{ in } GL\,(2.\ \mathbb{Z}_{11})$$

(b) Find the image of the elements 3 and 4 if

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 1 & & & 3 \end{pmatrix} \text{ is an old permutation.}$$

5 2

(c) Let G be a group. If $f : G \to G$ such that $f(x) = x^{-1}$ is a homomorphism, prove that G is Abelian.

(d) Give an example of ring elements a and b with the property that $ab = 0$ but $ba \neq 0$.

3. Answer the following questions : $5\times3=15$

(a) Prove that a group G is Abelian iff $(ab)^{-1} = a^{-1}b^{-1}$, for all $a, b \in G$.

(b) Show that the set of Gaussian integers $\mathbb{Z}[i] = \{a + ib \mid a, b \in \mathbb{Z}\}$ is a subring of the complex numbers $\mathbb{C}$.

OR

(c) Prove that a ring can have at most one unity.

(d) Define prime ideal. Show that $\langle 6 \rangle$ is not a prime ideal of $\mathbb{Z}_{36}$. Check whether $\langle 2 \rangle$ is a prime ideal of $\mathbb{Z}_{36}$.

$1+2+2=5$

OR

(e) Let ϕ be a homomorphism from a ring R to a ring S. Let A be a subring of R. then prove that $\phi(A) = \{\phi(a) \mid a \in A\}$ is a subring of S. What is $\phi(\ker\phi)$?

$4+1=5$

4. Answer ***either*** (a) ***or*** (b) part :

(a) (i) Show that the set $\mathbb{Z}_n = \{0, 1 \cdots n-1\}$, for $n \in \mathbb{N}$ is a group under addition modulo n. 5

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6.

(ii) Let $G = \left\{ \begin{bmatrix} a & a \\ a & a \end{bmatrix} \mid a \in \mathbb{R}, a \neq 0 \right\}$

Show that G is a group under matrix multiplication. 5

(b) Let $G = \langle a \rangle$ be a cyclic group of order n, then prove that $G = \langle a^K \rangle$ if and only if gcd $(K, n) = 1$. Hence find all the generators of $\mathbb{Z}_{30}$. 6+4=10

5. Answer *either* (a) *or* (b) part :

(a) State and prove the Lagrange's theorem. Give a condition for which the converse of the Lagrange's Theorem is true. Prove it. 1+5+4=10

(b) (i) If K is a normal subgroup of G and H is any subgroup of G then show that $H \cap K$ is a normal subgroup of G. 5

(ii) Prove that $Z(G)$ is a normal subgroup of G. 5

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6. Answer **either** (a) **or** (b) part :

(a) State and prove the fundamental theorem of group homomorphism. Hence show that any finite cyclic group of order n is isomorphic to the quotient group $\mathbb{Z}/N$, where $\langle \mathbb{Z}, + \rangle$ is the group of integers and $N = \langle n \rangle$ 1+5+4=10

(b) State and prove the Cayley's theorem. 10
